

Assignment 6

1. Show that $Z_4 \oplus Z_6 \simeq Z_2 \oplus Z_{12}$. (This was remaining from the earlier assignment)
2. Compute :
 - a) $\text{Aut}(\mathbb{Z}/p\mathbb{Z})$ for a prime p .
 - b) $\text{Aut}(\mathbb{Z}/8\mathbb{Z})$.
3. Show that the dihedral group of order 8 is not isomorphic to the Quaternions.
4. Compute the center of the group D_{2n} .
5. Let G be a group of order p^n and let H be a proper subgroup of G . Show that there exists $x \in G - H$ such that $xHx^{-1} = H$.